

CIRCE Winter School 2027

The Foundations of Extreme Value Analysis and Beyond

University of Padova, 25–26 January 2027
Via C. Battisti, 241, 35121 Padova (PD) – Room SC 60

1 Overview

How can we learn about rarely observed events, estimate risks beyond the available data, and understand extremes across several locations? Extreme value analysis provides a framework for addressing these questions.

Organised by the CIRCE network, the first edition of the winter school *The Foundation of Extreme Value Analysis and Beyond* will be hosted by the Department of Statistical Sciences at the University of Padova on 25–26 January 2027.

In this first edition, two complementary short courses, *The Foundation of Extreme Value Analysis* and *Modelling Spatial Extremes*, take participants from univariate extremes and tail inference to extremal dependence and spatial modelling. The school aims to:

- Build a solid understanding of extreme value theory and its statistical implications;
- Explain why extremes require inference methods tailored to distributional tails;
- Address key inferential challenges through established methods and selected recent developments;
- Explore extremal dependence and models for extremes observed across space.

Throughout the school, worked examples will help participants build intuition, explore theoretical results and understand the assumptions behind the methods. To complete this progression from theory to practice, real-data analyses from important applications will illustrate how to formulate questions about extremes, apply suitable methods and interpret results and uncertainty.

An *online reading group* will take place a few weeks after the school. Participants will study an assigned paper in small groups and present it at a virtual meeting, followed by a lecturer-led discussion and a Q&A session open to all school participants. This will offer an opportunity to explore selected topics further, practice critical reading and build research connections.

The school is intended for PhD students in statistics, mathematics, data science and related fields; master's students interested in pursuing a research career; and early-career postdoctoral researchers.

2 Foundations of Extreme Value Analysis

2.1 General information

Lecturers: Simone A. Padoan, Bocconi University
Stefano Rizzelli, University of Padova

Date: 25 January 2027, 9:30–18:30

Learning objectives. This course offers a concise yet rigorous introduction to univariate extreme value theory and inference for rare events. Starting from limit laws for sample maxima and threshold exceedances, it develops methods for tail inference, extreme quantile estimation and prediction of future extremes. Likelihood-based and Bayesian approaches will be discussed, with emphasis on the assumptions and inferential subtleties behind reliable conclusions.

Worked examples will clarify the theory and show how it connects with the main steps of practical inference. Real-data analyses from important applications will complete the picture, illustrating the interpretation of extreme risk estimates and their uncertainty.

2.2 Prerequisites

Master's-level knowledge of statistical inference (e.g., [3, 5]) and probability theory (e.g., [1], Chapter 5; [4], Appendices A1.1–A1.4). Familiarity with time series analysis (e.g., [2]) and stochastic processes (e.g., [6]) would also be helpful.

2.3 Course outline

The programme below is indicative and may be adapted according to the pace of the course and participants' needs. The outline below presents the planned topics; the depth of coverage may vary depending on the progress of the course.

1. Introduction to univariate extremes

- a. Domains of attraction, max-stable distributions and the block-maxima approach
- b. The generalised Pareto distribution, threshold stability and the peaks-over-threshold (POT) approach
- c. Extrapolation beyond the observed data: extreme quantiles and their estimation

2. Extremes of stationary sequences

- a. Mixing conditions, the extremal index and clusters of extremes

- b. Implications for statistical inference and applications
- 3. **Maximum likelihood inference for threshold exceedances**
 - a. The tail quantile process and second-order conditions
 - b. Rates of convergence, uniqueness and asymptotic normality of maximum likelihood estimators
 - c. Plug-in estimation of extreme quantiles
- 4. **Bayesian inference and prediction**
 - a. Priors, posteriors and posterior contraction rates; the Bernstein–von Mises theorem
 - b. Inference and prediction using the posterior tail predictive distribution

3 Modelling Spatial Extremes

3.1 General information

Lecturer: Carlo Gaetan, Ca' Foscari University of Venice

Date: 26 January 2027, 9:30–18:30

Learning objectives. Building on the first course, this course introduces the challenges of analysing extreme events observed across space. It explores how extremal dependence can be described and modelled, why marginal behaviour alone is insufficient to assess joint extremes, and how spatial models can support the study of risks affecting several locations.

Worked examples will help participants interpret patterns of extremal dependence and connect model assumptions to their practical implications. Analyses of real spatial data from important applications will bring these ideas together, illustrating model fitting, the interpretation of results and the assessment of uncertainty.

3.2 Prerequisites

Master's-level knowledge of probability and statistical inference, together with the main concepts introduced in *Foundations of Extreme Value Analysis*. Familiarity with stochastic processes would be helpful.

3.3 Course outline

The programme below is indicative and may be adapted according to the pace of the course and participants' needs. The outline below presents the planned topics; the depth of coverage may vary depending on the progress of the course.

1. **Introduction to model-based geostatistics**
 - a. Spatial random fields

- b. Gaussian random fields
- 2. **Spatial model with marginal extreme value distributions**
 - a. Hierarchical model
 - b. Gaussian anamorphosis
- 3. **Measuring the spatial dependence of extremes**
 - a. Dependence coefficients and tail asymptotics
- 4. **Asymptotic models for spatial extremes**
 - Definition of an extreme spatial events
 - Max-stable processes
 - Generalized Pareto processes
- 5. **Sub-asymptotic models for spatial extremes**
 - Random scale construction
 - Conditional models
- 6. **Estimation of spatial dependence models**
 - a. Likelihood-based estimation
 - b. Recent advances using likelihood-free neural-network-based estimators
- 7. **Case study**
 - a. Spatial modeling of meteorological variable

References

- [1] P. Billingsley. *Probability and Measure*. Wiley Series in Probability and Mathematical Statistics. John Wiley & Sons, New York, 3rd edition, 1995.
- [2] G. E. P. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung. *Time Series Analysis: Forecasting and Control*. John Wiley & Sons, Hoboken, New Jersey, 5th edition, 2015.
- [3] G. Casella and R. L. Berger. *Statistical inference*, volume 2. Duxbury Pacific Grove, CA, 2002.
- [4] P. Embrechts, C. Klüppelberg, and T. Mikosch. *Modelling Extremal Events: for Insurance and Finance*, volume 33. Springer Science & Business Media, Berlin, Heidelberg, 1997.
- [5] N. Mukhopadhyay. *Introductory Statistical Inference*. Statistics: A Series of Textbooks and Monographs. Chapman and Hall/CRC, 2006.
- [6] S. I. Resnick. *Adventures in Stochastic Processes*. Birkhäuser, Boston, 1992.